

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

06

10 questions · ~15 min

Q1

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The number a is 60% greater than the positive number b . The number c is 45% less than a . The number c is how many times b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

PROBLEM-SOLVING AND DATA ANALYSIS

The mean score of 8 players in a basketball game was 14.5 points. If the highest individual score is removed, the mean score of the remaining 7 players becomes 12 points. What was the highest score?

- A. 20
- B. 24
- C. 32
- D. 36

Q3

PROBLEM-SOLVING AND DATA ANALYSIS

Objects R and S each travel at a constant speed. The speed of object R is half the speed of object S. Object R travels a distance of $4x$ inches in y seconds. Which expression represents the time, in seconds, it takes object S to travel a distance of $24x$ inches?

- A. $12y$
- B. $3y$
- C. $16y$
- D. $6y$

	Amount invested	Balance increase
Account A	\$500	6% annual interest
Account B	\$1,000	\$25 per year

Two investments were made as shown in the table above. The interest in Account A is compounded once per year. Which of the following is true about the investments?

- A. Account A always earns more money per year than Account B.
- B. Account A always earns less money per year than Account B.
- C. Account A earns more money per year than Account B at first but eventually earns less money per year.
- D. Account A earns less money per year than Account B at first but eventually earns more money per year.

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A school district is forming a committee to discuss plans for the construction of a new high school. Of those invited to join the committee, 15% are parents of students, 45% are teachers from the current high school, 25% are school and district administrators, and the remaining 6 individuals are students. How many more teachers were invited to join the committee than school and district administrators?

STUDENT-PRODUCED RESPONSE

--	--	--	--

.	/	.	.
---	---	---	---

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Weight (pounds)	Frequency
13	12
14	8
15	5
16	7
17	9
18	10
19	13
20	7

The frequency table summarizes a data set of the weights, rounded to the nearest pound, of 71 tortoises. A weight of 39 pounds is added to the original data set, creating a new data set of the weights, rounded to the nearest pound, of 72 tortoises. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

- A. The mean of the new data set is greater than the mean of the original data set, and the median of the new data set is greater than the median of the original data set.
- B. The mean of the new data set is greater than the mean of the original data set, and the medians of the two data sets are equal.
- C. The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.
- D. The mean of the new data set is less than the mean of the original data set, and the medians of the two data sets are equal.

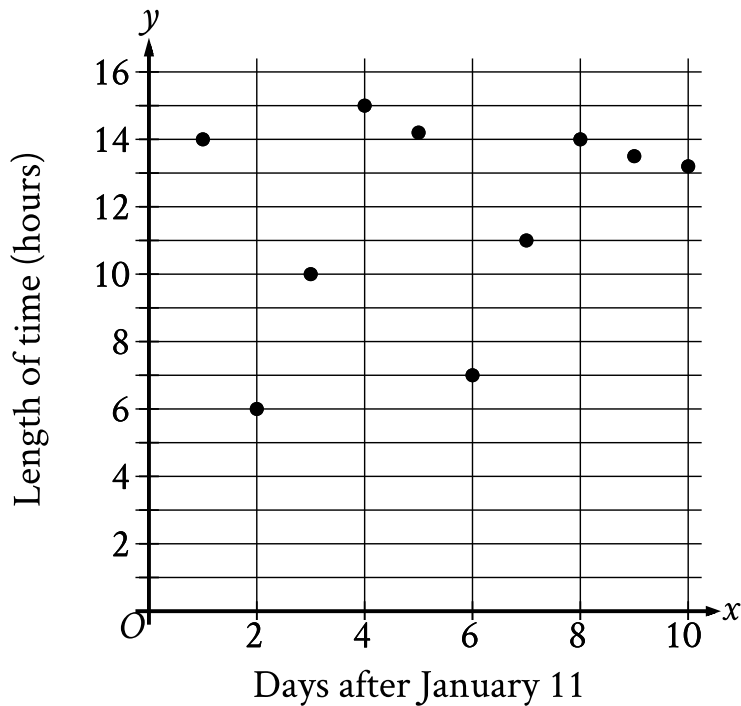
Q7

PROBLEM-SOLVING AND DATA ANALYSIS

A certain park has an area of 11,863,808 square yards. What is the area, in square miles, of this park? 1 mile = 1,760 yards

- A. 1.96
- B. 3.83
- C. 3,444.39
- D. 6,740.8

The scatterplot shows the relationship between the length of time y , in hours, a certain bird spent in flight and the number of days after January 11, x .



- The scatterplot has 10 data points.
- The data points are spread out.
- The data points have the following coordinates:
 - (1 comma 14)
 - (2 comma 6)
 - (3 comma 10)
 - (4 comma 15)
 - (5 comma 14.2)
 - (6 comma 7)
 - (7 comma 11)
 - (8 comma 14)
 - (9 comma 13.5)
 - (10 comma 13.2)

What is the average rate of change, in hours per day, of the length of time the bird spent in flight on January 13 to the length of time the bird spent in flight on January 15?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

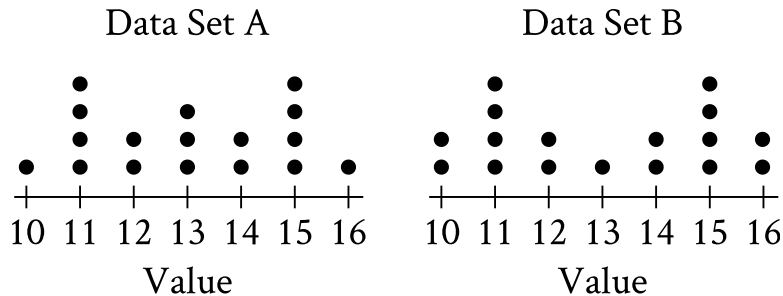
PROBLEM-SOLVING AND DATA ANALYSIS

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

10 / 12

The dot plots represent the distributions of values in data sets A and B.



- For the dot plot titled Data Set A:
 - The number line ranges from 10 to 16 in increments of 1.
 - The data for the dot plot are as follows:
 - 10: 1 dot
 - 11: 4 dots
 - 12: 2 dots
 - 13: 3 dots
 - 14: 2 dots
 - 15: 4 dots
 - 16: 1 dot
- For the dot plot titled Data Set B:
 - The number line ranges from 10 to 16 in increments of 1.
 - The data for the dot plot are as follows:
 - 10: 2 dots
 - 11: 4 dots
 - 12: 2 dots
 - 13: 1 dot
 - 14: 2 dots
 - 15: 4 dots
 - 16: 2 dots

Which of the following statements must be true?

- I. The median of data set A is equal to the median of data set B.

II. The standard deviation of data set A is equal to the standard deviation of data set B.

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II